Simulation of the Spread of Diseases with the Help of SIRD Modelling in Python

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Abstract—The aim of this paper was to perform an in-depth study on the SIRD epidemiological model, which offers a framework for analysing disease dynamics, and how it can be implemented in Python programming language. Epidemiological models have been used to study infectious diseases for a long time. Both, the SIRD and SIRDV model were implemented in python and tested using influenza statistics for the 2012 flu strain in the United States. The SIRD model uses four compartments susceptible, infected, recovered and deceased. It was verified using the calculated infection rate from real world data with a 2.4% error. The SIRDV model uses an added compartment: vaccinated. It was tested with three different vaccination rates (0.00121, 0.005, 0.05) to illustrate variations in how vaccines can prevent diseases. The inclusion of vaccines led to the susceptible population to falling rapidly. It was found that higher the vaccination rate, lower the susceptible population.

Keywords— SIRD model, SIRDV model, susceptible, infected, recovered, deceased, vaccinated, compartmental models, transmission rate, recovery rate, death rate, vaccination rate, vaccine efficacy.

I. Introduction

EPIDEMIOLOGICAL or Compartmental models are a basic modelling method that are often utilized to model the spread of diseases and their dynamics (Martcheva, 2015). Compartmental models have had an evident surge in their usage in recent years right around when the Covid-19 pandemic started. Epidemiologists employed such models to study and predict trends in the spread of corona virus, and how the disease could affect sects of the population. They also help in researching prevention methods and measures to take for the safety of a population.

At its core, an epidemic can be understood by mapping out who was infected and when. These models rely on the idea of classifying people into states or compartments, where an individual can only exist in one compartment at any given time. At the start of the transmission process, there is atleast one initial case. These conditions are set in the simplest compartmental model: the SIR model, where S represents Susceptible (individuals who are exposed to the disease), I represents Infected (individuals who have the disease and can spread it to others), and R represents Recovered (individuals who no longer have the disease).

This paper is going to discuss an extension on this model, called the SIRD model with the added parameter of D, which represents Deceased (individuals who passed away from the disease). If one person is initially infected, then the remainder of the population will become susceptible. Depending on how the disease spreads the susceptible will either stay susceptible or get infected and recover or pass away (Baude & Kimms, 2024).

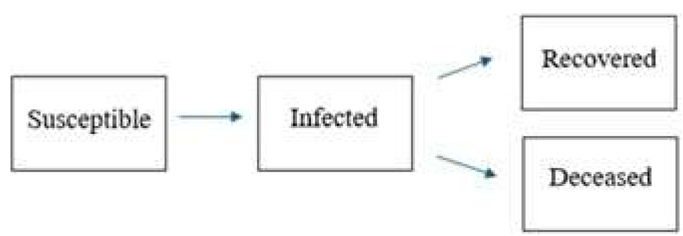

Fig. 1: movement of people in sird model

This model uses very simple differential equations that can be applied to different parameters. The parameters used are transmission rate, mortality rate, reproduction rate, recovery rate and total population. SIRD model will be tested along with another model: the SIRDV model, which has the added parameter of V, which represents Vaccinated (individuals who have received a vaccination). Those individuals will thereafter not be able to get infected.

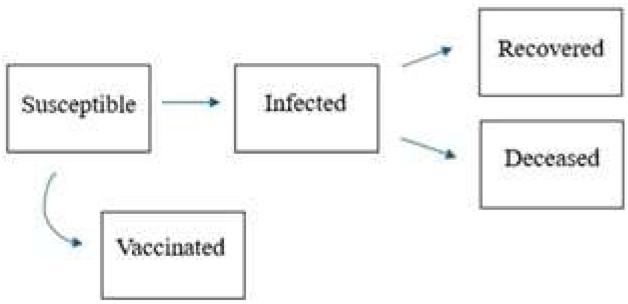

Fig. 2: movement of people in sirdv model

These models are generally applied for short-term forecasting of diseases, as they can realistically predict the sequence of an outbreak. In the case of long-term predictive analysis, it often necessitates assumption making, which can lead to very ambiguous results. This paper will also focus on how the equations of the disease models can be executed in a modified setting, depending on the parameters included in the modelling process.

A. Objectives

1) To explore the use of the compartmental models of epidemiology in simulating the transmission of diseases focusing on the SIRD model.
2) To identify how these models can be used in predicting the effects of such diseases on the population.
3) To demonstrate how the model can be implemented in the Python programming language.
4) To find accurate data sets regarding transmission rates, mortality rates, reproduction rates, incubation period and use them to test and simulate the SIRD and SIRDV model for the spread of the influenza A in the U.S.A for the year 2012.
5) To validate results for the SIRD model using the true values gathered from real world data.

II. Previous Works

Compartmental epidemiological models rely on the idea of classifying people into states or compartments, where an individual can only exist in one compartment at any given time. (Tolles & Luong, 2020). The original compartment model to study disease dynamics was published in a paper called "A Contribution to the Mathematical Theory of Epidemics.", with the hopes of introducing a system to understand the stream of infectious disease outbreaks that were widespread at the time. (Kermack & McKendrick, 1927)

“Simulation” refers to the graphical representation of how the disease spreads among the population based on equations crafted for the model. Epidemiological models can range from basic models like the SIR model to more complex or stochastic models that use computer simulations. (Xiao-Na Han, 2009). Usually, an infected person is introduced into a population, where people are susceptible to the given disease. Said disease will move through the population, where each affected individual will be removed from the infected population because of recovery or because of death. The total initial population is considered a constant for ease in modelling and is only modified regarding the number of deceased individuals.

An analysis of the SIR model for Covid-19, showed an increasing growth for the infection rates. Prohibiting migration and promoting isolation of the infected were concluded to be the safest ways to slow the spread of the disease (Din & Algehyne, 2021). The article was published before any vaccine had been officially created for Covid-19. An important note mentioned in the paper is to run the epidemiological model multiple times with different parameters to judge the spread of disease efficiently. A new model was created with fractional differential equations to comprehend the monkeypox virus (Peter, et al., 2022). Humans and rodents were considered as the population for this model. The parameters used in the compartmental model were established using reported data from Nigeria (Peter, et al., 2022). Results were extracted using different input parameters and studied, to provide a systematic model to manage monkeypox. This study hoped to be used as a basic guide for the steps to manage the monkeypox outbreak. (Peter, et al., 2022)

A study published by the University of Maine, adapted the SIR model to influenza datasets but there was an uncertainty for the flu vaccination rates within the research as some sources may have overestimated them. A lack of complete data could also lead to this. The model was verified to be accurate by making parameter values constant and making sure the results remained consistent (Dorr, 2019). Diseases are much more complex than what we can implement with these simple models, but it is still a competent way to study their dynamics (Skrip & Townsend, 2019). In addition to collecting more accurate datasets, using statistics can help in accurate estimation of parameter values (Dorr, 2019).

One of the most reliable methods of preventing and reducing transmittable diseases is vaccination (Statista, 2024). The model testing revealed that depending on the daily vaccination rate, there is a decrease in the number of deaths for each simulation. Vaccinations can also reduce the severity of illnesses (Centers for Disease Control and Prevention, 2024). Testing the model with the inclusion of vaccination rate can produce interesting results.

Such epidemiological models can offer information for planning prevention and control strategies, advocating for increased healthcare infrastructure, isolation of the infected, and appropriate tracking of infections. They help in designing and evaluating public health interventions, how they should be conducted and adjusted (Xiang, et al., 2021) .

The usage of computer simulation enhances the ease of epidemiological models. Computerized simulation would also have access to the same public data and would be aided by hypotheses to depict disease dynamics (Sofonea, Cauchemez, & Boëlle, 2022). It had the added advantage of being able to create visual representation of data on its own. This simplifies the process of understanding and potentially fixes any error in the method. The use of digital methods allow the user to quickly change data, initializations, or approaches.

Implementing in code also allows for the option of taking user input, leading to modifying the models via actual interaction. It is recommended for transparency in the choices made for including or excluding data sets (Orbann, Sattenspiel, Miller, & Dimka, 2017). This could make the results of the study be reproduced or replicated. Moreover, the use of computer simulations also introduces the possibility of deriving results on its own. It can also evaluate the results to determine safety measures and other calculated quantities to prevent epidemics. Interestingly, such compartmental models can also be applied to actual computer viruses (Kephart & White, 1999).

A. *The Working of the SIRD Model*

The SIRD model uses the following parameters:

Parameters	**Description**
S(t)	Susceptible population
I(t)	Infected population
R(t)	Recovered population
D(t)	Deceased population
β	Transmission rate
γ	Recovery rate
Μ	Death rate
N = S + I + R + D	Total population

(Din & Algehyne, 2021) (Ilekura, 2021)

The differential equations for the SIRD model are: (Bhattacharjee, et al., 2023)

$$\frac{dS}{dt} = -\beta\frac{\mathrm{SI}}{\mathrm{N}}$$

This equation gives the rate of change of susceptible population. The number of susceptible individuals, S, depends on the number of infected individuals, I, transmission rate, β and total population, N (Magri & Doan, 2020). The negative sign illustrates that susceptible population decreases overtime as more individuals get infected.

$$\frac{dI}{dt} = \beta\frac{\mathrm{SI}}{\mathrm{N}} - \gamma\mathrm{I} - \mu\mathrm{I}$$

This equation gives the rate of change of infected population (Magri & Doan, 2020). The number of infected individuals, I, is given by the number of infected individuals that recover, γI, and number of infected individuals that die, μI, subtracted from the total number of susceptible individuals getting infected, $\beta\frac{\mathrm{SI}}{\mathrm{N}}$.

$$\frac{dR}{dt} = \gamma\mathrm{I}$$

This equation gives the rate of change of recovered population (Magri & Doan, 2020). The number of recovered individuals, R, is given by the number of infected individuals that recover, γI, where γ is the recovery rate and I is the number of infected individuals.

$$\frac{dD}{dt} = \mu\mathrm{I}$$

This equation gives the rate of change of deceased population (Magri & Doan, 2020). The number of recovered individuals, R, is given by the number of infected individuals that die, μI, where μ is the death rate and I is the number of infected individuals.

The SIRD model also uses two other parameters, R0 and M, which are the reproduction rate of the disease and fatality rate for infect population respectively. These parameters, however, will not be included in the implementation as they are not contributing to the graphical results. (Dorr, 2019) (Ilekura, 2021)

B. *The Working of the SIRDV Model*

The SIRDV model is a simple expansion on the SIRD model. It includes all parameters from the SIRD model and a few new parameters that are mentioned below:

Parameters	**Description**
V(t)	Vaccinated population
α	Vaccination rate
e	Vaccine efficacy
N = S + I + R + D + V	Total population

The total population here, N, will have the inclusion of vaccinated population who are now protected from the disease. The differential equations for the SIRDV model are:

$$\frac{dS}{dt} = -\beta\frac{\mathrm{SI}}{\mathrm{N}} - \alpha\mathrm{S}$$

The number of susceptible people here also declines overtime but there is an added term of αS which corresponds to the number of susceptible people who are getting vaccinated at the rate of α.

These 3 equations are the same as those mentioned in the SIRD model. The infected, recovered and deceased compartments only depend on certain factors already discussed in the former model. (Manca, 2022) (Ilekura, 2021)

$$\frac{dV}{dt} = \alpha S - \beta \frac{eVI}{N}$$

This equation gives the rate of change (Magri & Doan, 2020) of the vaccinated population. The total number of vaccinated individuals is obtained from the $\beta \frac{eVI}{N}$ being subtracted from N the number of susceptible individuals getting vaccinated, αS. The $\beta \frac{eVI}{N}$ term represents the population who are vaccinated but can still get infected.

Vaccine efficacy or efficiency is given by e, which if equal to 1, means the vaccine is fully effective. If e is less than 1, then the vaccine offers incomplete protection against the disease. The SIRDV model can be simplified further by excluding the concept of vaccine efficacy. (Usherwood, LaJoie, & Srivastava, 2021) (Sebbagh, Bencheriet, & Kechida, 2024)

III. Research Methodology

A. Implementation of the Model through Python

The code will be using Euler's method to solve the differential equations. While most implementations of the model include the parameters R0 (basic reproduction number) and M (case fatality ratio), the produced one will not make use of them, as they don't directly affect the graphical representation. To create the model in python, the parameters have been represented as single letters for ease in understanding.

Parameters	Description	Representation in code
S(t)	Susceptible population	s
I(t)	Infected population	i
R(t)	Recovered population	r
D(t)	Deceased population	d
V(t)	Vaccinated population	v
β	Transmission rate	b
γ	Recovery rate	g
M	Death rate	m
α	Vaccination rate	a
e	Vaccine efficacy	e
N	Total population	N

The value of N will change as the value of D changes, as deceased individuals are no longer included in the total population. It will be initialized as 1000 for the models in python to show results on a small scale.

B. Database Searches

To test the models with real data, published records for transmission rates, mortality rates, incubation period and more for the United States of America were examined. World health Organization, The U.S. Census and Centers for Disease control and Prevention were chosen for seeking this data as they are trusted sources with interactive and weekly updating records. The data for the years 2012 will be used as it was a major flu strain in the United States (Centers for Disease Control and Prevention, 2023) (World Health Organization, n.d.). An error that may occur is the inaccuracy in the estimation of transmission rates, incubation rates, mortality rates, and other parameters for the influenza A. It is important to note that the CDC could have incomplete data or data which could be affected by other ILI's (Influenza Like Illnesses).

C. Estimation of Transmission rate, Recovery rate, and Death rate

Transmission rates were reported to be 8% or between 2% and 11% for the disease (Tokars, Olsen, & Reed, 2017). β value ranges from 0.4 to 0.6 so b will be set as 0.5 in the model. This means that every infected person has the possibility of transmitting the disease to half of a person on the susceptible population. Since the value is less than 1, it shows a lower transmission rate. (Magri & Doan, 2020)

The flu virus generally lasts from 3 to 7 days or even more if there are complications. Calculating the reciprocal of the number of days that the flu lasts would give the recovery rate. Taking the number of days as 7, γ will be set as 1/7 for the model. (Centers for Disease Control and Prevention, 2024)

Lastly the mortality rate can range anywhere from 0.01% to around 0.1% during harsh flu strains. The CDC also reports that the mortality rate for the flu is 1.8 per 100,000 people. Here, the value of μ will be set as 0.001 using the 0.1% statistic (Chowdhury, Islam, Hossain, Tabassum, & Peace, 2021) (Centers for Disease Control and Prevention, 2024).

D. Estimation of Vaccination rate and Vaccine efficacy

The code requires the rate of the population getting vaccinated every day (Forslid & Herzing, 2024). This data is not readily available as a single figure. Reports show that 26% of the 18- 49 age bracket, 42% of the 50-64 age bracket and 67% of the 65+ age bracket were vaccinated during the 2012 flu season. (Statista, 2020)

Using this, it can be calculated that 44.33% of the general adult population were vaccinated. The same adult population sat at 234,719,000. Dividing this percentage by the total number of days in 2012, which is 366 as it is a leap year, the everyday vaccination rate comes to 0.121%. This value would be set as the parameter α. (U.S. Census Bureau, 2012)

However, since the testing of influenza data is done on a much smaller scale of 1000 people, this is too small of a percentage for the population initialized in the testing. To better visualize the impact of vaccinations on a population, the value α or a in the code will be set with values 0.005 and 0.05.

a=0.005 means that 0.5% of the population gets vaccinated every day. This is not an entirely unrealistic value to be used as it would mean that the entire population would get vaccinated in just 200 days, which is typically how long a flu season lasts. The value of a= 0.05, meaning 5% of the population gets vaccinated every day (Forslid & Herzing, 2024), is to see how a higher vaccination rate will affect the graph.

The CDC Archive reports anywhere between 47% to 63% efficacy for the influenza A vaccine in the 2012-2013 flu strain. (CDC Archive, 2023) The efficacy is higher for people over the age of 65. The CDC archive also recounts the vaccine effectiveness to be a more specific 49% for all ages in the same season. (CDC Archive, 2018) (Centers for Disease Control and Prevention, 2024). 0.049 or 49% will be used as the value of ‘e’ in the SIRDV model.

IV. Results and Discussion

A. Analysis of Graphs in SIRD model

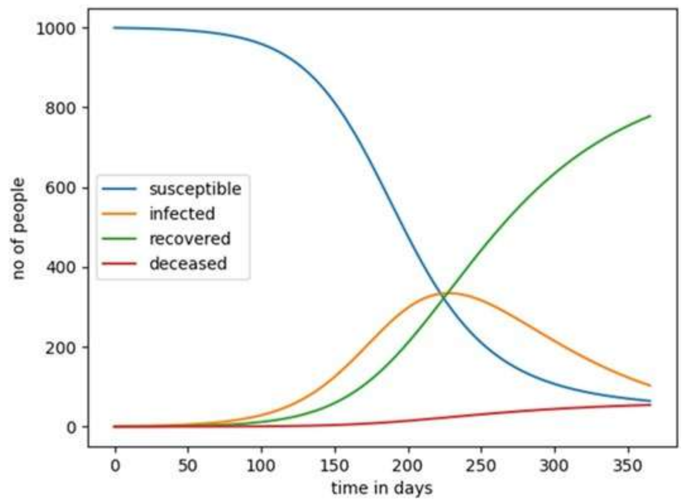

Fig. 3: graph generated for the SIRD model

From the above graph in Fig. 3, it is observed that the susceptible population dramatically plummets to 65 people at the end of the simulation. The infected population begins to surge, eventually reaching a peak of 334 people at day 228. At the end of the simulation there is a total of 105 infected people in the population. This value being calculated as a ratio of the total initial population, gives 0.105. Similarly, the ratio calculated from downloadable Flunet database for the same gives 0.1025 for the same year. (Global Influenza Surveillance & Response System, 2023).

There is only a 2.4% error in the observed ratio from the graph and the real-life data from the database. This shows accurateness in the testing which validates the above model and the parameter values chosen. The recovered population also gradually increases overtime. The deceased population is approximately 54 people when the simulation ends. The flu season typically peaks around December or winter months in general (Centers for Disease Control and Prevention, 2024), but since we consider a population of only 1000 people, this peak occurs very early on, compared to actual reports.

B. Analysis of Graphs in SIRDV Model

α=0.00121

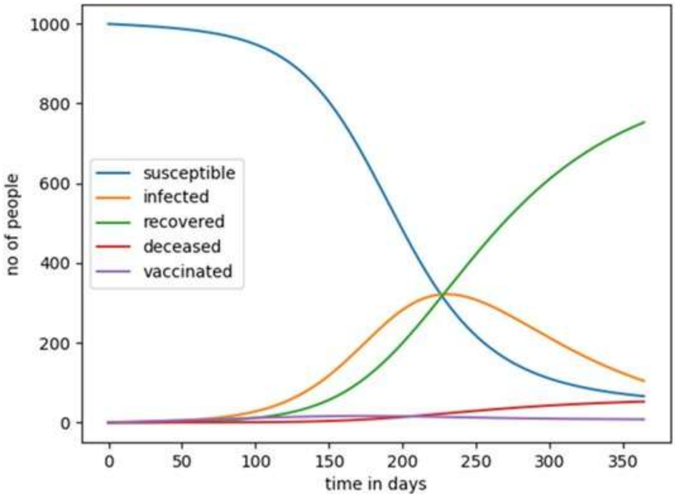

Fig. 4: SIRDV graph with a=0.00121

In Fig. 4, as discussed, the value of a=0.00121 is not high enough to see any significant changes for the population size being considered. The deceased, susceptible and infected populations at the end of the simulation is approximately the same as that of the previous graph. On day 226, susceptible, infected and recovered numbers are roughly the same. The vaccinated population at end sits at 8 people.

α=0.005

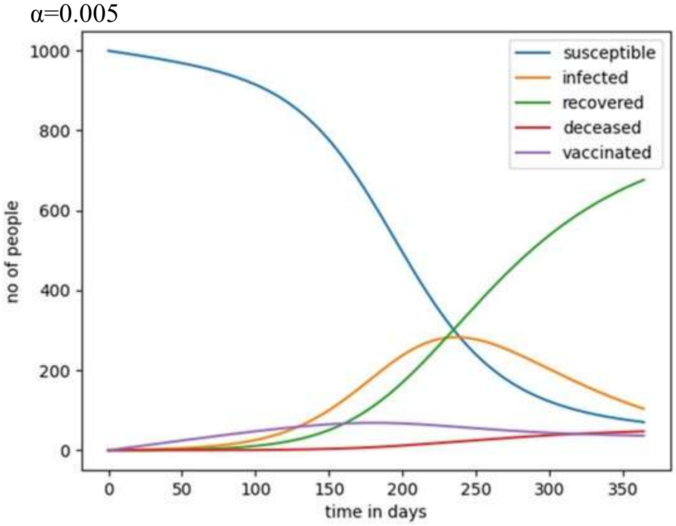

Fig. 5: SIRDV graph with a=0.005

With a=0.005, the graph in Fig. 5 for the SIRDV model is shows some differences. This figure demonstrates how vaccines can reduce exposure to and deaths from illnesses. As the rate of vaccination increases by 0.5% everyday, the susceptible population declines to 70 people, which is 7% of the total population. To reiterate, the vaccination rate assumed and not calculated estimation for better visual insight. The vaccinated population was highest at day 180 with 68 people getting vaccinated.

The deceased population starts to increase around day 83. It sits at 47 people at the end. As the considered population size is small, and because of the assumed vaccination rate, this amount is not very distinct from the previous graph. Recovered population also follows the same trends as in the previous graph. Infected population peaked ten days later compared to the previous graph, on day 237 with 282 people.

α=0.05

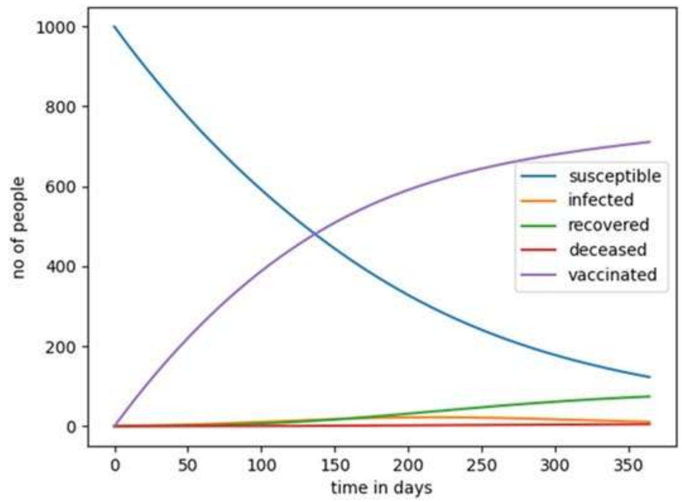

Figure 6: SIRDV graph with a=0.05

With the higher vaccination rate, the graph in Fig. 4 for the model is noticeably different. As the rate of vaccination increases by 5% everyday, the susceptible population declines to 12.3% of the total population. As is evident from the graph, only the vaccinated and susceptible population undergo major changes. The deceased population remains low throughout the simulation. It is almost stagnant, only 5 people out of the entire population pass away at the end. This number is almost tenth of the total deceased population from the SIRD graph. Recovered population showed a slight increase at the end whereas the quantity of infected people had a slight growth.

It is deduced how different vaccination rates affect the population. In summary, the higher the vaccination rate, the faster the population gets vaccinated, the lower the susceptible population. When similar simulations like this are performed, they produce results from which appropriate safety and control measures are developed to keep the population unharmed.

V. Conclusion and Future Works

This project explored the use of compartmental models in studying disease dynamics, delving into the working of two common models: SIRD and SIRDV. They are simple models predominantly used for short-term prediction of outbreaks. The SIRD implementation in python produced a 2.4% error when comparing observed and true rates. The SIRD implementation in python was tested with different daily vaccination rates to bring about diverse results. The outcomes of projects such as this one, could be analysed to update healthcare policies, interventions, and decision-making processes aimed at controlling infectious disease outbreaks in the real world.

The model heavily relies on some assumptions of parameters and a fixed population, which doesn't take varying birth rates, mortality rates or migration into account. The model also assumes that after recovering, an individual can't get affected again. A more complex model can be created by introducing the idea of immunity to a disease wearing off over time.

Taking an example of flu vaccines, the immunity to getting infected lasts for about 6 months, depending on their efficacy. New variants of the flu virus can develop by then, which is why it is recommended to retake flu shots to protect ourselves from new flu strains. By implementing this idea into the model, it would result in a sect of the population that can possibly get reinfected. Moreover, including geographical discrepancies, hospital capacity, and asymptomatic carriers can also make the model more conclusive.

Appendix

The code that was used in conducting the research is available at the GitHub page linked below. https://github.com/arthika333/sird-sirdv.git

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